

- 15.6** Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
<i>C</i>	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
<i>EXPER</i>	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
<i>EXPER</i> ²	-0.0033 (0.0011)	-0.0031 (0.0011)	-0.0012 (0.0011)	-0.0012 (0.0011)	-0.0023 (0.0007)
<i>SOUTH</i>	-0.2128 (0.0338)	-0.2384 (0.0344)	-0.3261 (0.1258)	-0.3261 (0.2495)	-0.2326 (0.0317)
<i>UNION</i>	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
<i>N</i>	716	716	1432	1432	1432

(standard errors in parentheses)

- a. The OLS estimates of the $\ln(WAGE)$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?

a.	OLS 1987	OLS 1988	Differ
EXPER	0.1270	0.1265	Almost the same
EXPER ²	-0.0033	-0.0031	Almost the same
SOUTH	-0.2128	-0.2384	Slightly different
UNION	0.1445	0.1105	Noticeably different

Assumption: Assume $u_i = 0$, i.e. there's no individual heterogeneity

- b. The $\ln(WAGE)$ equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}) \quad (XR15.6)$$

Explain any differences in assumptions between this model and the models in part (a).

The panel data model allows each individual to have their own intercept u_i , which relax the assumption we describe in part (a)

- c. Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?

	OLS 1987	OLS 1988	FE
EXPER	0.1270	0.1265	0.0575
EXPER ²	-0.0033	-0.0031	-0.0012
SOUTH	-0.2128	-0.2384	-0.3261
UNION	0.1445	0.1105	0.0822

The coefficient of EXPER shows the most difference.
(Percentage wise)

- d. The F -statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F -distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.

Numerator degrees of freedom: $N-1 = 716-1 = 715$

Denominator degrees of freedom: $NT-N-K_s = 716 \cdot 2 - 716 - 4 = 712$

The degree of freedom of F is: $(715, 712)$

The 1% level of significance critical value for $F_{(715, 712)}$ is 1.19

Since $11.68 > 1.19$, we reject the null hypothesis, i.e. there are individual differences.

- f. Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

	FE	RE
EXPER	0.0575	0.0986
EXPER ²	-0.0012	-0.0023
SOUTH	-0.3261	-0.2326
UNION	0.0822	0.1027

The coefficient of EXPER shows the most difference.

$$t = \frac{b_{FE,k} - b_{RE,k}}{[\widehat{\text{var}}(b_{FE,k}) - \widehat{\text{var}}(b_{RE,k})]^{1/2}} = \frac{b_{FE,k} - b_{RE,k}}{[\text{se}(b_{FE,k})^2 - \text{se}(b_{RE,k})^2]^{1/2}} \quad (15.36)$$

$$\text{By (15.36), } t_{\text{EXPER}} = \frac{0.0575 - 0.0986}{(0.033^2 - 0.022^2)^{1/2}} = -1.67$$

$$t_{\text{EXPER}^2} = \frac{-0.0012 - (-0.0023)}{(0.0011^2 - 0.0009^2)^{1/2}} = 1.30$$

$$t_{\text{SOUTH}} = \frac{-0.3261 - (-0.2326)}{(0.1258^2 - 0.0317^2)^{1/2}} = -0.77$$

$$t_{\text{UNION}} = \frac{0.0822 - 0.1027}{(0.0312^2 - 0.0245^2)^{1/2}} = -1.06$$

(We use column (3) because this is what the Hausman test uses the conventional se.)

$$(-1.67)^2 + (1.30)^2 + (-0.77)^2 + (-1.06)^2 = 6.1954,$$

which is smaller than $\chi^2(4)_{0.05} \approx 9.488$

So we fail to reject the Hausman test,

i.e. we can't reject "There's no endogeneity in the random effect model."

So the random effect model is appropriate.

15.20 This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

- a. Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?

```
> library(POE5Rdata)
> library(plm)      # Panel data models (fixed/random effects)
> library(lmtest)    # Hypothesis tests
> library(car)       # F-tests
>
> data(star)
>
> # =====
> # (a) OLS Regression (No Fixed or Random Effects)
> # =====
>
> ols_model <- lm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch,
+                 data = star)
> summary(ols_model)
```

Call:
lm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
freelunch, data = star)

Residuals:

	Min	1Q	Median	3Q	Max
	-107.220	-20.214	-3.935	14.339	185.956

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	437.76425	1.34622	325.180	< 2e-16 ***
small	5.82282	0.98933	5.886	4.19e-09 ***
aide	0.81784	0.95299	0.858	0.391
tchexper	0.49247	0.06956	7.080	1.61e-12 ***
boy	-6.15642	0.79613	-7.733	1.23e-14 ***
white_asian	3.90581	0.95361	4.096	4.26e-05 ***
freelunch	-14.77134	0.89025	-16.592	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 5759 degrees of freedom
(20 observations deleted due to missingness)
Multiple R-squared: 0.09685, Adjusted R-squared: 0.09591
F-statistic: 102.9 on 6 and 5759 DF, p-value: < 2.2e-16

a. From the OLS result:

The students perform better in reading when they're in small class, and it is statistically significant.

The students perform better in reading when there's a teacher aide, but it is not statistically significant.

The students perform better in reading if the teacher has more experience, and it is statistically significant.

The student's sex or race do make a difference: The student perform better in reading if the student isn't a boy or is white or asian, and both of them are statistically significant.

- b. Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [Hint: specify *SCHID* as the cross-section identifier and *ID* as the “time” identifier.]

```
> # =====
> # (b) School Fixed Effects Model
> # =====
>
> # Rename 'id' to 'stuid' to avoid conflict with plm's reserved index name 'id'
> star2 <- star
> star2$stuid <- star$id
> star2$id <- NULL
>
> # Create panel data object: cross-section = schid, "time" = stuid
> pdata <- pdata.frame(star2, index = c("schid", "stuid"))
>
> fe_model <- plm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch,
+               data = pdata,
+               model = "within", # Fixed Effects
+               effect = "individual")
>
> summary(fe_model)
Oneway (individual) effect Within Model

Call:
plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
     freelunch, data = pdata, effect = "individual", model = "within")

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Residuals:
    Min.   1st Qu.   Median   3rd Qu.    Max.
-102.6381 -16.7834  -2.8473  12.7591  198.4169

Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
small          6.490231   0.912962   7.1090 1.313e-12 ***
aide           0.996087   0.881693   1.1297  0.2586
tchexper       0.285567   0.070845   4.0309 5.629e-05 ***
boy            -5.455941   0.727589  -7.4987 7.440e-14 ***
white_asian    8.028019   1.535656   5.2277 1.777e-07 ***
freelunch     -14.593572   0.880006 -16.5835 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 4628000
Residual Sum of Squares: 4268900
R-Squared: 0.077592
Adj. R-Squared: 0.063954
F-statistic: 79.6471 on 6 and 5681 DF, p-value: < 2.22e-16

> # Compare OLS vs Fixed Effects coefficients
> cat("\n--- OLS Coefficients vs Fixed Effects Coefficients ---\n")

--- OLS Coefficients vs Fixed Effects Coefficients ---
> coef_compare <- cbind(
+   OLS = coef(ols_model)[-1], # Drop intercept
+   FE = coef(fe_model)
+ )
> print(round(coef_compare, 4))
```

	OLS	FE
small	5.8228	6.4902
aide	0.8178	0.9961
tchexper	0.4925	0.2856
boy	-6.1564	-5.4559
white_asian	3.9058	8.0280
freelunch	-14.7713	-14.5936

We can see that most of the coefficients don't change a lot, except that the coefficient of white_asian increases more than 100%.

- c. Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?

```
> # =====  
> # (c) Test for Significance of School Fixed Effects  
> # =====  
>  
> pFtest(fe_model, ols_model)
```

F test for individual effects

```
data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch  
F = 16.698, df1 = 78, df2 = 5681, p-value < 2.2e-16  
alternative hypothesis: significant effects
```

The p-value of the F-test is $< 2.2 \times 10^{-16}$, which is statistically significant. So we reject the null hypothesis and obtain that at least one school has a non-zero fixed effect.

Under the condition that the school fixed effect is uncorrelated to the explanatory variable, the inclusion of the school fixed effect will have little influence on the coefficient estimates of that explanatory variable.

- d. Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.

```
> # =====
> # (d) School Random Effects Model + LM Test
> # =====
>
> re_model <- plm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch,
+               data = pdata,
+               model = "random",
+               effect = "individual")
>
> summary(re_model)
Oneway (individual) effect Random Effect Model
(Swamy-Arora's transformation)
```

```
Call:
plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
     freelunch, data = pdata, effect = "individual", model = "random")
```

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Effects:

	var	std.dev	share
idiosyncratic	751.43	27.41	0.829
individual	155.31	12.46	0.171

theta:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6470	0.7225	0.7523	0.7541	0.7831	0.8153

Residuals:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-97.4830	-17.2361	-3.2822	0.0372	12.8027	192.3460

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept)	436.126774	2.064782	211.2217	< 2.2e-16 ***
small	6.458722	0.912548	7.0777	1.466e-12 ***
aide	0.992146	0.881159	1.1260	0.2602
tchexper	0.302679	0.070292	4.3060	1.662e-05 ***
boy	-5.512081	0.727639	-7.5753	3.583e-14 ***
white_asian	7.350477	1.431376	5.1353	2.818e-07 ***
freelunch	-14.584332	0.874676	-16.6740	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 6158000
Residual Sum of Squares: 4332100
R-Squared: 0.29655
Adj. R-Squared: 0.29582
Chisq: 493.205 on 6 DF, p-value: < 2.22e-16

```
> # Compare coefficients across all three models
> cat("\n--- Coefficient Comparison: OLS vs FE vs RE ---\n")
```

--- Coefficient Comparison: OLS vs FE vs RE ---

```
> coef_all <- cbind(
+   OLS = coef(ols_model)[-1],
+   FE = coef(fe_model),
+   RE = coef(re_model)[-1] # Drop intercept
+ )
> print(round(coef_all, 4))
```

	OLS	FE	RE
small	5.8228	6.4902	6.4587
aide	0.8178	0.9961	0.9921
tchexper	0.4925	0.2856	0.3027
boy	-6.1564	-5.4559	-5.5121
white_asian	3.9058	8.0280	7.3505
freelunch	-14.7713	-14.5936	-14.5843

```
>
> # LM test: H0 = no random effects (Breusch-Pagan LM test)
> plmtest(re_model, type = "bp")
```

Lagrange Multiplier Test - (Breusch-Pagan)

```
data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch
chisq = 6677.4, df = 1, p-value < 2.2e-16
alternative hypothesis: significant effects
```

The coefficient of white_asian has the most difference, so it might be correlated to the school effects.

The LM test rejects the null hypothesis, so the school effects exist.

- e. Using the t -test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?

```
> # =====
> # (e) Hausman Test (FE vs RE) - Using t-statistic from Eq. (15.36)
> # =====
>
> hausman_test <- phptest(fe_model, re_model)
> print(hausman_test)
```

Hausman Test

data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch
 chisq = 13.809, df = 6, p-value = 0.03184
 alternative hypothesis: one model is inconsistent

```
>
> vars_of_interest <- c("small", "aide", "tchexper", "white_asian", "freelunch")
>
> b_fe <- coef(fe_model)[vars_of_interest]
> b_re <- coef(re_model)[vars_of_interest]
> se_fe <- sqrt(diag(vcov(fe_model)))[vars_of_interest]
> se_re <- sqrt(diag(vcov(re_model)))[vars_of_interest]
>
> t_stat <- (b_fe - b_re) / sqrt(se_fe^2 - se_re^2)
>
> cat("\n--- t-statistics (FE - RE), 5% significance level, critical value ≈ ±1.96 ---\n")
```

--- t-statistics (FE - RE), 5% significance level, critical value ≈ ±1.96 ---

```
> result_e <- data.frame(
+   b_FE = round(b_fe, 4),
+   b_RE = round(b_re, 4),
+   t_stat = round(t_stat, 4),
+   reject_H0 = abs(t_stat) > 1.96
+ )
> print(result_e)
```

	b_FE	b_RE	t_stat	reject_H0
small	6.4902	6.4587	1.1460	FALSE
aide	0.9961	0.9921	0.1284	FALSE
tchexper	0.2856	0.3027	-1.9377	FALSE
white_asian	8.0280	7.3505	1.2181	FALSE
freelunch	-14.5936	-14.5843	-0.0956	FALSE

The Hausmann test shows that we can't reject the null hypothesis for each of these four variables, i.e. we can't reject that the random effects and the explanatory variable are uncorrelated.

```
> # Apply the test separately to BOY
> b_fe_boy <- coef(fe_model)["boy"]
> b_re_boy <- coef(re_model)["boy"]
> se_fe_boy <- sqrt(vcov(fe_model)["boy", "boy"])
> se_re_boy <- sqrt(vcov(re_model)["boy", "boy"])
>
> t_boy <- (b_fe_boy - b_re_boy) / sqrt(se_fe_boy^2 - se_re_boy^2)
```

Warning message:
 In sqrt(se_fe_boy^2 - se_re_boy^2) : NaNs produced

```
> cat("\n--- t-statistic for BOY ---\n")
```

--- t-statistic for BOY ---

```
> cat("t =", round(t_boy, 4), "\n")
t = NaN
```

This may be due to that $FE \approx RE$ for this variable *BOY*.

Thus, we should use the fixed effect model by the Hausmann test.

- f. Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

```
> # =====
> # (F) Mundlak Test
> # =====
>
> mundlak_vars <- c("small", "aide", "tchexper", "boy", "white_asian", "freelunch")
>
> school_means <- aggregate(star2[, mundlak_vars],
+                           by = list(schid = star2$schid),
+                           FUN = mean)
> colnames(school_means)[-1] <- paste0(mundlak_vars, "_mean")
>
> star_mundlak <- merge(star2, school_means, by = "schid")
>
> mundlak_model <- lm(
+   readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
+   small_mean + aide_mean + tchexper_mean +
+   boy_mean + white_asian_mean + freelunch_mean,
+   data = star_mundlak
+ )
>
> summary(mundlak_model)
```

Call:
lm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
freelunch + small_mean + aide_mean + tchexper_mean + boy_mean +
white_asian_mean + freelunch_mean, data = star_mundlak)

Residuals:

	Min	1Q	Median	3Q	Max
	-106.579	-19.802	-3.851	14.826	191.128

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	464.07945	5.47058	84.832	< 2e-16 ***
small	6.56177	1.00367	6.538	6.79e-11 ***
aide	1.09202	0.96794	1.128	0.259289
tchexper	0.29460	0.07807	3.774	0.000162 ***
boy	-5.40859	0.79969	-6.763	1.48e-11 ***
white_asian	8.19637	1.68549	4.863	1.19e-06 ***
freelunch	-14.64249	0.96940	-15.105	< 2e-16 ***
small_mean	-22.55156	5.82335	-3.873	0.000109 ***
aide_mean	10.88891	5.36079	2.031	0.042280 *
tchexper_mean	0.82536	0.18108	4.558	5.28e-06 ***
boy_mean	-51.23797	7.16245	-7.154	9.51e-13 ***
white_asian_mean	-7.02301	2.35217	-2.986	0.002841 **
freelunch_mean	-3.62902	2.53316	-1.433	0.152025

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.96 on 5689 degrees of freedom
(84 observations deleted due to missingness)
Multiple R-squared: 0.117, Adjusted R-squared: 0.1151
F-statistic: 62.81 on 12 and 5689 DF, p-value: < 2.2e-16

```
> # H0: all mean coefficients = 0 (no correlation with unobserved heterogeneity)
> mean_vars <- c("small_mean", "aide_mean", "tchexper_mean",
+               "boy_mean", "white_asian_mean", "freelunch_mean")
>
> cat("\n--- Mundlak Test: Joint Significance of School-Level Means ---\n")

--- Mundlak Test: Joint Significance of School-Level Means ---
> linearHypothesis(mundlak_model,
+                 hypothesis.matrix = mean_vars,
+                 test = "Chisq")

Linear hypothesis test:
small_mean = 0
aide_mean = 0
tchexper_mean = 0
boy_mean = 0
white_asian_mean = 0
freelunch_mean = 0

Model 1: restricted model
Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
small_mean + aide_mean + tchexper_mean + boy_mean + white_asian_mean +
freelunch_mean

Res.Df    RSS Df Sum of Sq  Chisq Pr(>Chisq)
1   5695 5220109
2   5689 5105264   6    114845 127.98 < 2.2e-16 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> cat("\n--- Individual coefficients on school-level means ---\n")

--- Individual coefficients on school-level means ---
> mean_coefs <- summary(mundlak_model)$coefficients[mean_vars, ]
> print(round(mean_coefs, 4))

Estimate Std. Error t value Pr(>|t|)
small_mean -22.5516    5.8233 -3.8726  0.0001
aide_mean  10.8889    5.3608  2.0312  0.0423
tchexper_mean  0.8254    0.1811  4.5579  0.0000
boy_mean -51.2380    7.1625 -7.1537  0.0000
white_asian_mean -7.0230    2.3522 -2.9858  0.0028
freelunch_mean -3.6290    2.5332 -1.4326  0.1520
```

The Mundlak test has a very small p-value, which strongly reject the random effects assumption.

The school-average of all variables except FREELUNCH are statistically significant, so these variables are correlated with unobserved school-level heterogeneity.